Coding & Solution

Date	04 July 2026	Team ID	—
Project Name	India's Agriculture Crop Production Analysis	Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/saichaturya018/India-Agriculture-Crop-Production-Analysis
Programming Language(s)	Python, JavaScript
Framework(s) Used	Django, Pandas, Plotly.js, Bootstrap
Key Features Implemented	Data collection from government portals, data preprocessing & cleaning, crop production analysis by state, season, and year, trend analysis, interactive dashboards with charts and maps, export reports (CSV/PDF).
Pending / Incomplete Features	Real-time crop yield prediction model integration, satellite imagery analysis module, mobile app version.
Setup / Run Instructions	1. Clone the repository. 2. Run `pip install -r requirements.txt` for backend dependencies. 3. Run `npm install` inside the frontend directory. 4. Configure database settings in `.env` file. 5. Run the backend server: `python manage.py runserver`. 6. Run the frontend: `npm start`. 7. Open the application in browser at `http://localhost:3000`.

Code Quality Checklist

INDIA'S AGRICULTURE

CROP PRODUCTION ANALYSIS

Empowering Farmers. Enriching Agriculture. Ensuring Food Security.

SKILL WALLET

A SmartBridge Product

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	**Yes**
2	Meaningful variable and function names are used	**Yes**
3	Code includes comments / documentation where necessary	**Yes**
4	Error handling is implemented for critical operations	**Yes**
5	The application runs without critical errors	**Yes**
6	Code is committed to a version control repository (e.g., GitHub / GitLab)	**Yes**
7	Data validation is performed for user inputs and external data sources	**Yes**
8	Security best practices (authentication, authorization, input sanitization) are followed	**Yes**

Additional Notes / Comments:

The submission includes a well-structured architecture, modular and reusable components, and complete functional modules. Data pipelines, analytics services, and dashboards are working as expected. Version control commits effectively track all development changes. Follows coding standards and security practices. Excellent alignment with project objectives and requirements for India's Agriculture Crop Production Analysis.